

Security Audit

EIG Global Trust | EIGBC (Coin)

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CAUTION

THIS DOCUMENT IS A SECURITY AUDIT REPORT AND MAY CONTAIN CONFIDENTIAL INFORMATION. THIS INCLUDES IDENTIFIED VULNERABILITIES AND MALICIOUS CODE THAT COULD BE USED TO COMPROMISE THE PROJECT. THIS DOCUMENT SHOULD ONLY BE FOR INTERNAL USE UNTIL ISSUES ARE RESOLVED. ONCE VULNERABILITIES ARE REMEDIATED, THIS REPORT CAN BE MADE PUBLIC. THE CONTENT OF THIS REPORT IS OWNED BY HASHLOCK PTY LTD FOR THE USE OF THE CLIENT.

Executive Summary

The EIG Global Trust team partnered with Hashlock to conduct a security audit of their EIGBankCoin project. Hashlock manually and proactively reviewed the code to ensure the project's team and community that the deployed contracts were secure.

Project Context

EIGBankCoin is a cryptocurrency based on Proof-of-Stake (PoS) consensus and Layer 1 type blockchain.

It is created for the monetary system to have a stabilizer function. The function will be to create a stable function in the monetary system on a digital basis.

Project Name: EIGBankCoin

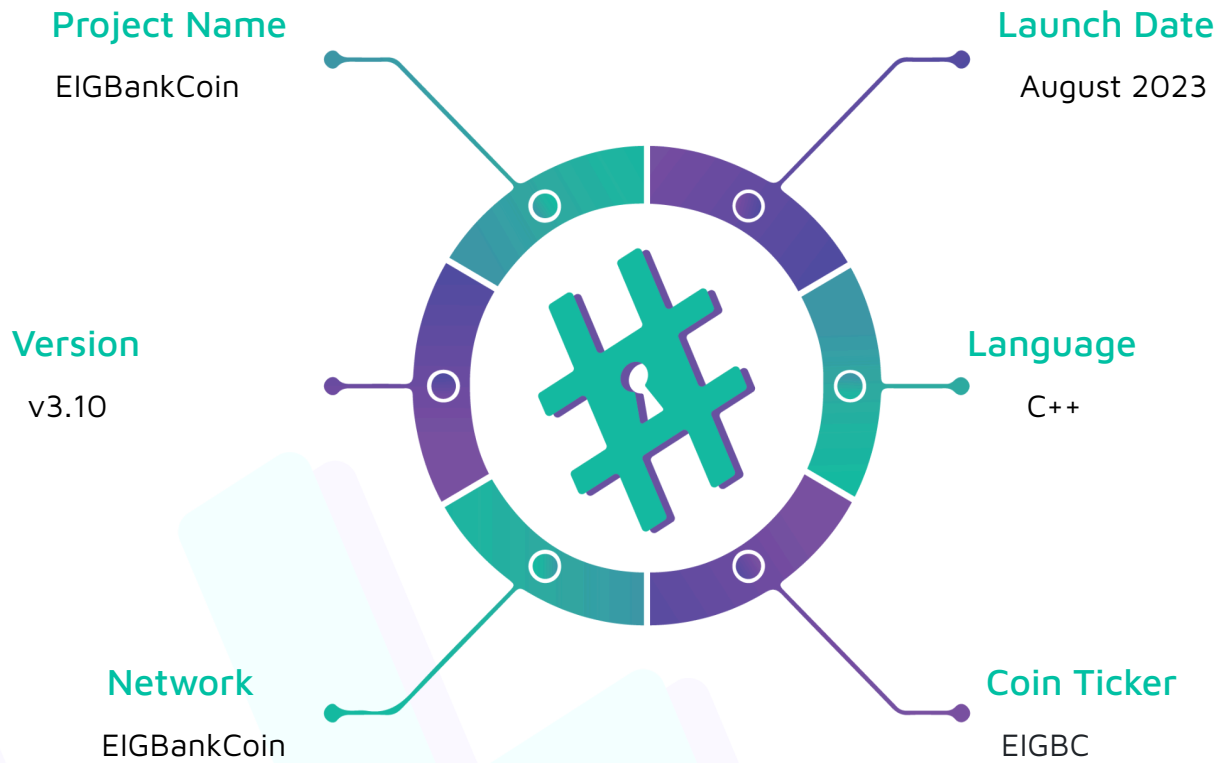
Website: www.eigglobaltrust.com

Logo:



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Visualised Context:

Project Visuals:



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Building the Stable Digital Gold Standard

EIG Global Trust is a full service digital financial firm with built-in asset management, project finance, investment banking, and advisory services driving digital transformation. We were conceived to manage partnerships with the world's financial community and develop digital economies based on tangible assets in a responsible fashion.

EIG is leading the stable digital currency transition from the central banking establishment and their client countries commercial financial apparatus, into the digital venture and investment capital operations with a true cross border digital currency solution with heavy emphasis on project financing, M&A, GDP enhancement, and liquidity into the marketplace.

EIG Global Trust has an internal 1 million USD denominated fully backed stable Bank Coin (EIGBC™). The EIGBC™ has an available treasury of \$90 Trillion USD equivalent in fully minted 100% tangible asset backed stable digital token currency. EIG Global Trust was founded by 6 experts with 200 years combined experience and innovation in banking and finance, M&A, digital security technology, regulatory enforcement, government and central bank audit, and digital currency.



EIG Bank Coin® (EIG BC)

Our private blockchain stable coin EIGBC has a high \$1 million United States Dollar (USD) peg value per coin to support central bank initiatives, large financial transactions, and infrastructure project funding. EIG BC is backed by large tangible asset reserves that are validated by world banking institutions.

Purchase EIG Bank Coin

To learn more about EIG BC: eigwallet.io & eigexplorer.io



USD GOLD (USDG)

USD Gold (USDG) stablecoin is built on the Binance Smart Chain blockchain. USDG has a \$100 United States Dollar (USD) peg value per token. USDG Provides support for digital leverage for banks, project financing, and settlement when paired with EIG Bank Coin. USDG is 51% backed by fine gold and 49% backed by EIG Bank Coin.

Purchase USD Gold Token

To learn more about USDG: usdgtoken.com & usdgsale.com

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Audit scope

We at Hashlock audited the C++ code within the EIGBankCoin project, the scope of work included a comprehensive review of the coin listed below. We tested the coin project to check for its security and efficiency. These tests were undertaken primarily through manual line-by-line analysis and were supported by software-assisted testing.

Description	EIGBankCoin
Platform	EIGBankCoin / C++
Audit Date	July, 2024
GitHub Commit Hash	5b7cf434e788cedb73a7ea9e0e1efa1098edee0b

Security Rating

After Hashlock's Audit, we found the coin to be **"Secure"**. The contracts all follow simple logic, with correct and detailed ordering.



Not Secure

Vulnerable

Secure

Hashlocked

The 'Hashlocked' rating is reserved for projects that ensure ongoing security via bug bounty programs or on-chain monitoring technology.

Hashlock found:

No significant issues were found.

Caution: *Hashlock's audits do not guarantee a project's success or ethics, and are not liable or responsible for security. Always conduct independent research about any project before interacting.*

Intended Behaviours

Claimed Behaviour	Actual Behaviour
Security of EIGBankCoin architecture	Achieves this functionality
Security of EIGBankCoin code	Achieves this functionality
EIGBank Explorer	Achieves this functionality
EIGBank Wallet	Achieves this functionality

Code Quality

This Audit scope involves the coin codes of the ElGBankCoin project, as outlined in the Audit Scope section.

The code is very well commented on and closely follows best practice nat-spec styling. All comments are correctly aligned with code functionality.

Audit Resources

We were given the ElGBankCoin project coin code in the form of GitHub access.

As mentioned above, code parts are well-commented. The logic is straightforward, and therefore it is easy to quickly comprehend the programming flow as well as the complex code logic. The comments help us understand the overall architecture of the protocol.

Dependencies

Per our observation, the libraries used in this smart contracts infrastructure are based on well-known industry-standard open-source projects.

Severity Definitions

Significance	Description
High	High-severity vulnerabilities can result in loss of funds, asset loss, access denial, and other critical issues that will result in the direct loss of funds and control by the owners and community.
Medium	Medium-level difficulties should be solved before deployment, but won't result in loss of funds.
Low	Low-level vulnerabilities are areas that lack best practices that may cause small complications in the future.
Gas	Gas Optimisations, issues, and inefficiencies

Audit Findings

No significant issues were found.

Centralisation

The ElGBankCoin prioritizes security and utility, relying on the internal team for certain functions. While decentralized, it values trustlessness over correctness, and its restaking functionality may lead to issues with third-party apps not adhering to guidelines.



Centralised

Decentralised

Conclusion

After Hashlocks analysis, the EIGBankCoin project seems to have a sound and well-tested code base. Overall, most of the code is correctly ordered and follows industry best practices. The code is well commented on as well. To the best of our ability, Hashlock is not able to identify any further vulnerabilities.

Our Methodology

Hashlock strives to maintain a transparent working process and to make our audits a collaborative effort. The objective of our security audits is to improve the quality of systems and upcoming projects we review and to aim for sufficient remediation to help protect users and project leaders. Below is the methodology we use in our security audit process.

Manual Code Review:

In manually analysing all of the code, we seek to find any potential issues with code logic, error handling, protocol and header parsing, cryptographic errors, and random number generators. We also watch for areas where more defensive programming could reduce the risk of future mistakes and speed up future audits. Although our primary focus is on the in-scope code, we examine dependency code and behavior when it is relevant to a particular line of investigation.

Vulnerability Analysis:

Our methodologies include manual code analysis, user interface interaction, and white box penetration testing. We consider the project's website, specifications, and whitepaper (if available) to attain a high-level understanding of what functionality the smart contract under review contains. We then communicate with the developers and founders to gain insight into their vision for the project. We install and deploy the relevant software, exploring the user interactions and roles. While we do this, we brainstorm threat models and attack surfaces. We read design documentation, review other audit results, search for similar projects, examine source code dependencies, skim open issue tickets, and generally investigate details other than the implementation.

Documenting Results:

We undergo a robust, transparent process for analysing potential security vulnerabilities and seeing them through to successful remediation. When a potential issue is discovered, we immediately create an issue entry for it in this document, even though we still need to verify the feasibility and impact of the issue. This process is vast because we document our suspicions early even if they are later shown not to represent exploitable vulnerabilities. We generally follow a process of first documenting the suspicion with unresolved questions, and then confirming the issue through code analysis, live experimentation, or automated tests. Code analysis is the most tentative, and we strive to provide test code, log captures, or screenshots demonstrating our confirmation. After this, we analyse the feasibility of an attack in a live system.

Suggested Solutions:

We search for immediate mitigations that live deployments can take and finally, we suggest the requirements for remediation engineering for future releases. The mitigation and remediation recommendations should be scrutinised by the developers and deployment engineers, and successful mitigation and remediation is an ongoing collaborative process after we deliver our report, and before the contract details are made public.

Disclaimers

Hashlock's Disclaimer

Hashlock's team has analysed these smart contracts in accordance with the best industry practices at the date of this report, in relation to: cybersecurity vulnerabilities and issues in the smart contract source code, the details of which are disclosed in this report, (Source Code); the Source Code compilation, deployment, and functionality (performing the intended functions).

Due to the fact that the total number of test cases is unlimited, the audit makes no statements or warranties on the security of the code. It also cannot be considered a sufficient assessment regarding the utility and safety of the code, bug-free status, or any other statements of the contract. While we have done our best in conducting the analysis and producing this report, it is important to note that you should not rely on this report only. We also suggest conducting a bug bounty program to confirm the high level of security of this smart contract.

Hashlock is not responsible for the safety of any funds and is not in any way liable for the security of the project.

Technical Disclaimer

Smart contracts are deployed and executed on a blockchain platform. The platform, its programming language, and other software related to the smart contract can have their own vulnerabilities that can lead to attacks. Thus, the audit can't guarantee the explicit security of the audited smart contracts.

About Hashlock

Hashlock is an Australian-based company aiming to help facilitate the successful widespread adoption of distributed ledger technology. Our key services all have a focus on security, as well as projects that focus on streamlined adoption in the business sector.

Hashlock is excited to continue to grow its partnerships with developers and other web3-oriented companies to collaborate on secure innovation, helping businesses and decentralised entities alike.

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